

Unit 1: Internet Basics

Aligned with WDP122S Syllabus

1. Communication Using the Internet

Basics of Computer Networks

A computer network is a group of interconnected computers and devices that can communicate and share resources (like printers or internet access). Networks form the foundation of all modern digital communication.

LAN and WAN

- **LAN (Local Area Network):** Connects computers within a limited geographical area, such as a home, school, or office building.
- **WAN (Wide Area Network):** Covers a large geographical area, often connecting multiple LANs across cities, countries, or the globe. The Internet is the largest example of a WAN.

OSI and TCP Network Models

Network models provide a theoretical framework for how data travels over a network.

- **TCP/IP Model:** The practical, 4-layer framework that powers the modern Internet (Application, Transport, Internet, Network Interface).
- **OSI Model:** A 7-layer theoretical reference model (Application, Presentation, Session, Transport, Network, Data Link, Physical) used to understand and standardize network architecture.

Concept of Internet, Applications, and Knowing the Internet

The **Internet** is a global "network of networks" that connects billions of devices using standard communication protocols.

Applications of the Internet are vast and include: the World Wide Web (information sharing), Email, File Transfer, Online Gaming, E-commerce, Video Conferencing, and Social Media.

Connecting to the Internet & What is an ISP?

To connect to the Internet, you require an **ISP (Internet Service Provider)**. An ISP is a company (like Comcast, AT&T, or a local telecom) that provides you with access to the broader internet. Connections are typically made via Wi-Fi, Ethernet, broadband (DSL/Cable), or cellular networks (4G/5G).

2. WWW and Web Browsers

World Wide Web (WWW)

The WWW is an information system running over the Internet where documents and web resources are linked together using hypertext. While the Internet is the physical infrastructure, the Web is the service you interact with.

Web Browsing Software's & Search Engines

- **Web Browsers:** Software applications (like Google Chrome, Mozilla Firefox, Apple Safari, or Microsoft Edge) used to access, retrieve, and view websites.
- **Search Engines:** Specialized websites (like Google or Bing) that index the web and help users find specific information based on keywords.

Understanding URL, Domain Name, and IP Address

- **URL (Uniform Resource Locator):** The complete web address used to find a specific page or resource (e.g., <https://www.example.com/page1>).
- **Domain Name:** The human-readable part of a network address that represents a website (e.g., example.com).
- **IP Address:** The unique numerical identifier assigned to every device connected to the Internet (e.g., 192.168.1.1). The Domain Name System (DNS) translates Domain Names into these IP addresses behind the scenes.

Using e-Governance Websites

e-Governance involves utilizing the Internet to deliver government services and information directly to citizens. This allows users to easily apply for passports, pay taxes, or check public records online securely.

3. Communications and Collaboration

Basics of Electronic Mail

Electronic Mail (Email) is a method of exchanging digital messages between people over the Internet using electronic devices.

Getting an Account, Sending/Receiving, and Using Emails

- **Getting an Email Account:** Users sign up with email service providers like Gmail, Yahoo, or Outlook to get a unique, personal address (e.g., username@domain.com).
- **Sending and Receiving:** To send an email, you compose a message, enter the recipient's address, and hit send. The email provider routes it across the Internet to the recipient's inbox, where they can open and read it.
- **Accessing Sent Emails:** All standard email clients feature a "Sent" or "Sent Mail" folder that automatically stores copies of all outgoing messages for your records and future reference.

Document Collaboration & Instant Messaging

- **Document Collaboration:** Using online platforms (like Google Docs or Microsoft 365) that allow multiple users to edit, comment on, and view the same document simultaneously from different locations.
- **Instant Messaging:** Real-time, text-based communication over the Internet (e.g., WhatsApp, Telegram, or Slack).

Netiquettes

Netiquette (Network Etiquette) refers to the unwritten rules of acceptable, polite, and professional behavior when communicating online. This includes avoiding typing in all caps (which looks like shouting), respecting privacy, and being concise.